

In our empirical analysis we provide estimates for both the traditional and the Anderson-van Wincoop gravity equations using alternative estimation methods. We show that, in practice, heteroskedasticity is quantitatively and qualitatively important in the gravity equation, even when controlling for fixed effects. Hence, we recommend to estimate the augmented gravity equation in levels, using the proposed PML estimator, which also adequately deals with the zero observations.

3. Constant-Elasticity Models

Despite their immense popularity, there are still important econometric flaws in empirical studies involving gravity equations. These flaws are not exclusive of this literature, and extend to many areas where constant-elasticity models are used. This section examines how the deterministic multiplicative models suggested by economic theory can be used in empirical studies.

In their non-stochastic form, the relationship between the multiplicative constant-elasticity model and its log-linear additive formulation is trivial. The problem, of course, is that economic relations do not hold with the accuracy of the physical laws. All that can be expected is that they hold on average. Indeed, here we interpret economic models like the gravity equation as the expected value of the variable of interest, $y \geq 0$, for a given value of the explanatory variables, x (see Goldberger, 1991, p. 5). That is, if economic theory suggests that y and x are linked by a constant-elasticity model of the form $y_i = \exp(x_i\beta)$, the function $\exp(x_i\beta)$ is interpreted as the conditional expectation of y_i given x , denoted $E[y_i|x]$.⁵ For example, using the notation in the previous sec-

⁵Notice that if $\exp(x_i\beta)$ is interpreted as describing the conditional median of y_i (or other conditional quantile) rather than the conditional expectation, estimates of the elasticities of interest can be obtained estimating the log-linear model using the appropriate quantile regression estimator (Koenker and Bassett, 1978). However, interpreting $\exp(x_i\beta)$ as a conditional median is problematic when y_i has a large mass of zero observations, like in trade data. Indeed, in this case the conditional median of y_i will be a discontinuous function of the regressors, which is generally not compatible with the standard economic theory.

tion, the multiplicative gravity relationship can be written as the exponential function $\exp[\ln(\alpha_0) + \alpha_1 \ln(Y_i) + \alpha_2 \ln(Y_j) + \alpha_3 \ln(D_{ij})]$, which is interpreted as the conditional expectation $E(T_{ij}|Y_i, Y_j, D_{ij})$.

Since the relation $y_i = \exp(x_i\beta)$ holds on average, but not for each i , there is an error term associated with each observation, which is defined as $\varepsilon_i = y_i - E[y_i|x]$.⁶ Therefore, the stochastic model can be formulated as

$$y_i = \exp(x_i\beta) + \varepsilon_i, \quad (6)$$

with $y_i \geq 0$ and $E[\varepsilon_i|x] = 0$.

As we mentioned before, the standard practice of log-linearizing equation (6) and estimating β by OLS is inappropriate for a number of reasons. First of all, y_i can be zero, in which case log-linearization is unfeasible. Second, even if all observations of y_i are strictly positive, the expected value of the log-linearized error will in general depend on the covariates and hence OLS will be inconsistent. To see the point more clearly, notice that equation (6) can be expressed as

$$y_i = \exp(x_i\beta) \eta_i,$$

with $\eta_i = 1 + \varepsilon_i / \exp(x_i\beta)$ and $E[\eta_i|x] = 1$. Assuming for the moment that y_i is positive, the model can be made linear in the parameters by taking logarithms of both sides of the equation, leading to

$$\ln(y_i) = x_i\beta + \ln(\eta_i). \quad (7)$$

To obtain a consistent estimator of the slope parameters in equation (6) estimating (7) by OLS, it is necessary that $E[\ln(\eta_i)|x]$ does not depend on x_i .⁷ Since $\eta_i = 1 + \varepsilon_i / \exp(x_i\beta)$, this condition is met only if ε_i can be written as $\varepsilon_i = \exp(x_i\beta) \nu_i$, where ν_i is a random variable statistically independent of x_i . In this case, $\eta_i = \nu_i$ and therefore is statistically independent of x_i , implying that $E[\ln(\eta_i)|x]$ is constant. Thus, the log-linear

⁶Whether the error term enters additively or multiplicatively is irrelevant for our purposes, as explained below.

⁷Consistent estimation of the intercept would also require $E[\ln(\eta_i)|x] = 0$.

representation of the constant-elasticity model is only useful as a device to estimate the parameters of interest under very specific conditions on the error term.

When η_i is statistically independent of x_i , the conditional variance of y_i (and ε_i) is proportional to $\exp(2x_i\beta)$. Although economic theory generally does not provide any information on the variance of ε_i , we can infer some of its properties from the characteristics of the data. Because y_i is non-negative, when $E[y_i|x]$ approaches zero, the probability of y_i being positive must also approach zero. This implies that $V[y_i|x]$, the conditional variance of y_i , tends to vanish as $E[y_i|x]$ passes to zero.⁸ On the other hand, when the expected value y is far away from its lower bound, it is possible to observe large deviations from the conditional mean in either direction, leading to greater dispersion. Thus, in practice, ε_i will generally be heteroskedastic and its variance will depend on $\exp(x_i\beta)$, but there is no reason to assume that $V[y_i|x]$ is proportional to $\exp(2x_i\beta)$. Therefore, in general, regressing $\ln(y_i)$ on x_i by OLS will lead to inconsistent estimates of β .

It may be surprising that the pattern of heteroskedasticity and, indeed, the form of all higher-order moments of the conditional distribution of the error term can affect the consistency of an estimator, rather than just its efficiency. The reason is that the non-linear transformation of the dependent variable in equation (7) changes the properties of the error term in a non-trivial way since the conditional expectation of $\ln(\eta_i)$ depends on the shape of the conditional distribution of η_i . Hence, unless very strong restrictions are imposed on the form of this distribution, it is not possible to recover information about the conditional expectation of y_i from the conditional mean of $\ln(y_i)$ simply because $\ln(\eta_i)$ is correlated with the regressors. Nevertheless, estimating (7) by OLS will produce consistent estimates of the parameters of $E[\ln(y_i)|x]$ as long as $E[\ln(y_i)|x]$ is a linear function of

⁸In the case of trade data, when $E[y_i|x]$ is close to its lower bound (i.e., for pairs of small and distant countries), it is unlikely that large values of trade are observed since they cannot be offset by equally large deviations in the opposite direction simply because trade cannot be negative. Therefore, for these observations, dispersion around the mean tends to be small.

the regressors.⁹ The problem is that these parameters may not permit identification of the parameters of $E[y_i|x]$.

In short, even assuming that all observations on y_i are positive, it is not advisable to estimate β from the log-linear model. Instead, the non-linear model has to be estimated.

3.1 Estimation

Although most empirical studies use the log-linear form of the constant-elasticity model, some authors (see Frankel and Wei, 1993, for an example in the international trade literature) have estimated multiplicative models using non-linear least squares (NLS), which is an asymptotically valid estimator for (6). However, the NLS estimator can be very inefficient in this context as it ignores the heteroskedasticity that, as discussed before, is characteristic of this type of model.

The NLS estimator of β is defined by

$$\hat{\beta} = \arg \min_b \sum_{i=1}^n [y_i - \exp(x_i b)]^2,$$

which implies the following set of first order conditions:

$$\sum_{i=1}^n \left[y_i - \exp(x_i \hat{\beta}) \right] \exp(x_i \hat{\beta}) x_i = 0. \quad (8)$$

These equations give more weight to observations where $\exp(x_i \hat{\beta})$ is large because that is where the curvature of the conditional expectation is more pronounced. However, these are generally also the observations with larger variance, which implies that NLS gives more weight to noisier observations. Thus, this estimator may be very inefficient, depending heavily on a small number of observations.

If the form of $V[y_i|x]$ was known, this problem could be eliminated using a weighted-NLS estimator. However, in practice, all we know about $V[y_i|x]$ is that, in general, it goes to zero as $E[y_i|x]$ passes to zero. Therefore, an optimal weighted-NLS estimator

⁹When $E[\ln(y_i)|x]$ is not a linear function of the regressors, estimating (7) by OLS will produce consistent estimates of the parameters of the best linear approximation to $E[\ln(y_i)|x]$ (see Goldberger, 1991, p. 53).